Milestone 1: Orintation & Data Loading

By

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HotelRevAI

Abstract

HotelRevAI is a data-driven project designed to analyze hotel booking data and generate revenue insights using Power BI and AI-based techniques. The main objective of this project is to help hotels understand booking patterns, seasonal demand, cancellations, and pricing trends in order to improve overall revenue performance.

The project uses a real-world hotel booking dataset collected from Kaggle (2018–2020). The data is imported into Power BI, where it is cleaned, transformed, and structured using data modeling techniques. Important attributes such as booking date, room type, number of guests, average daily rate (ADR), and cancellation status are used for revenue analysis.

Problem Statement

- Hotels holds large booking dataset but for using it effectively they need skilled strategist.
- Manual analysis leads to:
    - Incorrect pricing decisions.
    - Revenue loss during low-demand periods.
    - Missed profit opportunities during peak seasons.
- There is a need for an automated, data-driven revenue analysis system.

Project Idea

- Use hotel booking dataset to:
    - Understand booking behavior.
    - Identify revenue patterns.
    - Analyze seasonal demand and cancellations.
- Using AI, logical calculations and predictions for revenue

Proposed Solution

- An AI-driven system that:
    - Analyzes hotel booking and revenue data.
    - Identifies trends and patterns.
    - Supports smarter pricing and planning decisions.
- Helps hotels:
    - Improve occupancy rate.
    - Reduce cancellations.
    - Maximize overall revenue.

Data Collection

- **Source:** Kaggle
- **Dataset:** Hotel Booking Demand
- **Time Period:** 2018 – 2020

Process Flow

Step 1: Raw Hotel Dataset (CSV)

The process begins with the raw hotel booking dataset collected from Kaggle. This dataset contains booking records such as arrival dates, room type, number of guests, average daily rate, and cancellation status. At this stage, the data is unprocessed and may contain missing or inconsistent values.

Step 2: Data Cleaning & Preprocessing

In this step, the raw data is cleaned and prepared. This includes Removing missing or null values, correcting incorrect data types, removing duplicate entries, filtering irrelevant information. This step ensures that the data is accurate and reliable for analysis.

Step 3: Feature Selection

After cleaning, important revenue-related features are selected. These include booking date, room type, number of guests, Average Daily Rate (ADR), cancellation status, total nights, and revenue. Only the relevant attributes are retained to focus on revenue analysis.

Step 4: Structured Data Model

Next, the cleaned and selected data is organized into a structured data model. The data is arranged Year-wise, Month-wise, By room type, By booking status. This structured model makes analysis easier and supports AI-based calculations.

Step 5: Revenue Analysis & AI Insights

Finally, the structured data is used for revenue analysis. Using Power BI and AI features, the system Identifies revenue trends, Detects seasonal demand patterns, Analyzes cancellation impact, Generates forecasting insights. This step converts raw data into meaningful business intelligence.

Conclusion

The orientation phase helped in clearly understanding the project objective, problem statement, and overall scope of HotelRevAI. During this stage, the goal of developing an AI-driven revenue analysis system for hotels was defined, along with identifying the required tools and dataset. The project direction, workflow, and expected outcomes were finalized to ensure a structured approach.

In the data loading phase, the hotel booking dataset was successfully collected from Kaggle and imported into Power BI. The dataset was verified, structured, and prepared for further processing. Data types were checked, key revenue-related fields were identified, and the dataset was made ready for modeling.